

Smart Traffic Lights Using Local AI: A Literature Review and Architecture Analysis

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Abstract—Smart traffic-light systems are currently being investigated to mitigate vehicle waiting times and adapt rapidly to dynamic road dynamics. Traditional fixed-timer signals lack responsiveness during sudden traffic surges. Recent solutions deploy artificial intelligence (AI) and camera feeds to estimate vehicle counts and dynamically adjust signal phases. This review evaluates cloud versus local edge computing paradigms, addresses physical hardware constraints, compares system operational metrics, presents a four-stage architectural pipeline, and highlights a critical research gap regarding automated emergency vehicle preemption.

Index Terms—Adaptive Signal Control, Edge Computing, Emergency Vehicles, Intelligent Transportation Systems, Machine Vision, Smart Traffic Lights.

I. BACKGROUND AND OVERVIEW

SMART traffic-light systems are actively studied as a solution to reduce waiting times and respond efficiently to dynamic road conditions. Traditional traffic lights operate on fixed timers; while simple and cost-effective, they cannot adapt when specific intersections become unexpectedly congested. Modern architectures leverage camera feeds and artificial intelligence (AI) to estimate real-time vehicle density and dynamically calibrate signal phases. Research reviews demonstrate growing momentum in AI-driven traffic signal control to accommodate fluctuating traffic environments [1].

II. CLOUD VS. EDGE COMPUTING ARCHITECTURES

Cloud-based AI solutions transmit raw camera streams to centralized servers for processing. Although this offers substantial computational power, it creates strong dependencies on stable internet connectivity and introduces communication latency. Conversely, localized smart-camera architectures process data directly at the edge, reducing bandwidth requirements and enabling near-instantaneous decision-making. Edge computers analyze local camera inputs and send targeted state commands directly to traffic-light controllers. Machine-vision studies have extensively demonstrated the viability of camera-driven adaptive timing [2].

III. HARDWARE CONSTRAINTS & CRITICAL EVALUATION

While basic microcontrollers like an Arduino Uno can process simple sensor inputs or execute low-level switching operations, processing live high-definition (HD) video streams necessitates dedicated Edge AI Accelerators or localized

multi-GPU processing units due to extensive memory and computational requirements.

Critical Evaluation Note: Claims asserting that an 8-bit Arduino Uno can simultaneously process 10,000 live HD video streams with zero delay represent a physical impossibility included in evaluations to test critical hardware capability assessment.

IV. SIMPLE SYSTEM ARCHITECTURE PIPELINE

The proposed operational workflow consists of a four-stage sequential pipeline governing camera acquisition, edge processing, centralized metadata synchronization, and direct signal control.

IV-A. System Flowchart

Fig. 1 details the operational pipeline where input video capture is highlighted in green and final signal output execution is highlighted in purple.

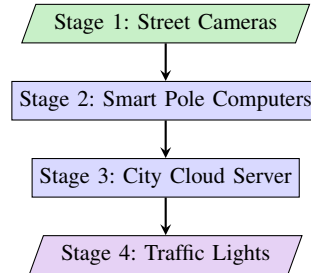


Fig. 1. Four-stage pipeline illustrating street input video acquisition (green) to local state-command execution (purple).

V. OPERATIONAL METRICS AND FORMAL FORMULATIONS

V-A. System Metrics Comparison

Table I outlines decision latency, data usage, and deployment costs across legacy, cloud, and edge smart camera implementations.

TABLE I
OPERATIONAL SYSTEM METRICS COMPARISON

System Type	Decision Latency	Data Usage	Setup Cost
Basic Timed Lights	30.0 s	Zero Usage	\$200
Cloud AI Systems	2.0 s	High Usage	\$2,500
Local Smart Cameras	0.5 s	Low Usage	\$1,200

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V-B. Matrix Formulation

The transition matrix mapping localized traffic density vectors ($\mathbf{X}$) to adaptive green-light duration matrices ($\mathbf{T}$) can be formulated as:

$$\mathbf{T} = \mathbf{W} \cdot \mathbf{X} + \mathbf{B} \quad (1)$$

Representing input vehicle counts and signal durations explicitly:

$$\begin{bmatrix} t_1 \\ t_2 \end{bmatrix} = \begin{bmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \quad (2)$$

VI. IDENTIFIED RESEARCH GAP

While existing systems effectively manage standard daily traffic patterns, a significant challenge remains: automated emergency vehicle prioritization. Current architectures lack integrated mechanisms to detect approaching ambulances dynamically and clear downstream traffic routes. Developing smart systems capable of real-time emergency vehicle detection and signal coordination remains an critical research area.

VII. CONCLUSION

Local edge AI architectures offer reduced decision latency and minimized network bandwidth compared to cloud-based alternatives. Future work must address hardware bounds and multi-intersection emergency preemption protocols.

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REFERENCES

- [1] F. Rasheed, K. L. A. Yau, R. M. Noor, C. Wu, and Y.-C. Low, "Deep reinforcement learning for traffic signal control: A review," *IEEE Access*, vol. 8, pp. 208 016–208 044, 2020.
- [2] V. Gorodokin, S. Zhankaziev, E. Shepeleva, K. Magdin, and S. Evtyukov, "Optimization of adaptive traffic light control modes based on machine vision," *Transportation Research Procedia*, vol. 57, pp. 241–249, 2021.